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Study protocol for a cross-sectional online survey investigating patient preferences and experiences of waiting for elective cardiac surgery

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ABSTRACT

Introduction Being on a waiting list for elective (planned) cardiac surgery can be physically and psychologically challenging for patients. Research suggests that stress associated with waiting for surgery is dependent on different individual and contextual factors. However, most data on patients' experiences of waiting for surgery and preferences for waiting list management derives from non-cardiac clinical populations. The aim of the current study is to explore patients' experiences of being on a waiting list for elective cardiac surgery, and their views on how the waiting experience could be improved in the future. This work will inform the patient management strategy during the waiting period for surgery across the four major hospitals in London directly involved in this study, and potentially beyond by transferring learning to other services.

Methods and analysis This is a mixed-methods study that will collect quantitative and qualitative data using a cross-sectional online survey. Patients who are on waiting lists for elective surgery across four major cardiac surgery departments in London hospitals, and are at least 18 years old, will be invited by their healthcare team via text message or letter to complete the survey. The target sample size of non-randomly selected participants will be 268. Bivariable and multivariable regression models will be used to assess associations between survey items measuring the impact of the cardiac condition on specific life domains (eg, daily activities, social and family relationships, hobbies, sexual life), anxiety and depression symptoms as measured by the Patient Health Questionnaire-4 and survey items evaluating experiences of health services. Data on experience and preferences for improvements to the waiting experience will be analysed with qualitative content analysis using an inductive approach.

Ethics and dissemination This study was reviewed and granted ethical approval by the East of England—East Cambridge Research Ethics Committee. Findings from this study will be disseminated through peer-reviewed journals, a research website and social media and with an online event engaging patients, members of the public, healthcare professionals and other relevant stakeholders.

STRENGTHS AND LIMITATIONS OF THIS STUDY

⇒ This study will include data from a large pool of patients planned to have cardiac surgery at multiple hospitals in London.
⇒ The online survey will support efforts to reach a large number of patients within a short time frame.
⇒ The cross-sectional design will limit the exploration of causality between variables.
⇒ Data from some groups of participants (eg, elderly, disabled and non-digital groups) might not be captured in this study as a result of an online survey and/or limited access to resources (eg, internet connection, computer or smart phone access).
⇒ A convenience sample could lead to selection bias.

Trial registration numb NCT05996640

INTRODUCTION

Cardiovascular diseases (CVDs; that is, coronary heart disease, stroke, other heart and arteriole diseases, hypertensive diseases and other diseases of the circulatory system) represent a leading cause of death worldwide, and are the second leading cause of death in the UK, with the mortality rate being around 29%.[1] It is estimated that CVD care in the National Health Service (NHS) in England costs a total of £7.4 billion per year, with costs to the wider economy going up to £15.8 billion.[2] Reducing waiting times for cardiac surgery is paramount for the NHS, especially following the onset of the COVID-19 pandemic. Indeed, the proportion of patients waiting more than 18 weeks for cardiac surgery across England sharply rose from about 10% in the first months of 2020 to about 40% in June to July 2020, and stabilised to around 20% in the first half of 2021 (British Heart Foundation report[3]). Recent data from

February 2023 showed that in the NHS in England 33.3% of patients waiting for elective cardiac surgery are on the waiting list for more than 18 weeks.[4] Data from a cohort of 1099 patients across seven UK centres exploring the effect of delays (due to the COVID-19 pandemic) in elective cardiac surgery, found that, at 1 year, of the 916 patients meeting primary outcomes (ie, surgery, percutaneous therapy, death), 92% underwent surgery (after a median waiting time of 195 days), 3% had a percutaneous intervention, 3% declined surgery and 1% died. Although mortality risk might seem relatively low, authors highlighted that this could be associated with adverse outcomes, such as increased anxiety and overall worsening of quality of life due to a longer wait.[5] Results from a meta-analysis (22 studies and more than 66 000 patients across Europe, New Zealand, North and South America) showed that rates per 1000 patient-weeks were of 1.1 for death and 1.0 for non-fatal myocardial infarction (MI) for coronary artery bypass grafts (CABG) patients, and 0.1 for death and 0.4 for MI for percutaneous coronary intervention patients.[6]

Beyond the above-mentioned serious adverse effects of waiting, there is evidence suggesting that being on a waiting list negatively impacts on physical functioning[7] and that there is a high association between the deterioration of physical functioning and anxiety.[8] In this regard, it is important to carefully consider the psychological experience of being on the waiting list. Indeed, anxiety and depression, along with poor quality of life, are often reported among patients waiting for surgery.[9] Data from cardiac surgery populations suggests that the level of depressive symptoms might be lower compared with anxiety. The latter is reported by almost all patients and it reaches a clinical level in almost one-third of them.[10 11] Inconsistent findings emerged about the association between anxiety and sex with some studies showing that being female is a predictor of higher levels of anxiety,[12 13] while others do not.[10 11] In terms of issues from which anxiety arises, it emerged that patients were more anxious around knowledge of surgery and anaesthesia which highlights the importance of establishing proper communication while waiting for surgery.[11] Indeed, a review focusing on the psychological aspects associated with waiting (mainly transplant and cancer surgery) found that regular communication and updates about their waiting status along with acknowledgement of the emotional burden experienced by patients, and peer support would be helpful to reduce distress.[14] Moreover, previous data on patients waiting for orthopaedic or cardiac surgery suggested that the perception and quality of waiting are not necessarily associated with its length.[15 16] Indeed, it was found that factors such as restriction of functioning and uncertainty associated with lack of control (due to being on a waiting list) were negatively experienced by patients waiting for orthopaedic or cardiac surgery, regardless of the duration of waiting.[15] Yet, other aspects of waiting were less distressing and even perceived positively, for example, when patients were able to accept waiting, perceiving it as part of life and when they developed coping strategies (eg, use of distraction, downward comparison with patients in worse situations and use of social/family support).[15] In some cases, waiting was also considered an opportunity to be more prepared for the surgery and to make good use of their time.[15] However, there is data highlighting that postponement and/or cancellation of cardiac surgery could induce negative emotions like anxiety and fear.[17] Also, Gagliardi and colleagues[14] found that the impact of being on a waiting list is more detrimental for specific groups such as women, people of younger age and those who cope through avoidance, venting or emotional rather than task-oriented strategies.[14] Other studies suggested that both internal (ie, personality) and external factors (ie, attention received, participation in decision-making for care) could influence patients' experiences. Similarly, factors that are considered as negative by patients (ie, dissatisfaction with the healthcare organisation and lack of a supportive social network) deserve attention from the clinical team during the pre-surgery period.[18] Also, due to the high incidence of anxiety and depressive symptoms in people suffering from CVDs,[17 19–22] it is also crucial to try to mitigate the negative impact that these risk factors could have on the cardiac pathophysiology.

Policy approaches for reducing waiting times in healthcare, reviewed by Kreindler[23] included government policies and identified three main strategies that were used internationally: increasing supply (eg, pay for increased activity and staff capacity from other countries and/or the private sector); reducing demand (eg, giving priority to patients meeting certain urgency thresholds); and influencing local organisations' behaviours (eg, setting mandatory targets with direct incentives, encouraging competition).[23] While some of these institutional-level policies can reduce waiting times, they do not necessarily reflect patient-level preferences or improve the experience of waiting. These strategies do not consider or address the psychological distress that patients experience while waiting for surgery. A more holistic approach combining policy interventions with the patients' perspective on how to improve waiting time management might be more effective in enhancing care quality and satisfaction with services. In particular, the patient's experiences may depend not only on the duration of waiting time, but also on other factors such as personal health and care experiences, attitudes towards waiting, coping skills and support from and stable communication with the healthcare team. Current peer-reviewed evidence on these factors largely concerns orthopaedic or patients with cancer,[14–16] with a lack of published data on cardiac patients' specific experiences and preferences.

This study aims to fill this gap in knowledge by investigating the experiences of patients who are waiting for cardiac surgery, and their preferences in terms of cardiac surgery across four major cardiac surgery hospitals in London. Specifically, the aims of this study will be to investigate:

1. Patients' experiences of how their cardiac condition impacts their quality of life and what they experience being on a waiting list for cardiac surgery.
2. Patients' preferences in relation to the delivery of the surgery (eg, whether patients would prefer to wait longer to be operated on by a specific surgeon or have their surgery sooner by any available surgeon (that are equally qualified and skilled)) and their views on improving the waiting process.

The overall aim of this project will be to provide key stakeholders (eg, clinicians, waiting list managers, medical directors, policymakers) with an in-depth understanding of patients' experiences and preferences about being on a waiting list for elective cardiac surgery. Specifically, findings from this project will contribute to a better understanding of (1) patients' experiences of waiting, by for instance, providing a greater awareness of potential risk factors (eg, higher or lower levels of anxiety associated with specific patients' characteristics) that could impact on the cardiac condition; and of (2) patients' preferences for managing the waiting list in the future, thus supporting the design of more effective waiting list tailored to the preferences and experiences of patients. This will inform a wider quality improvement project aimed at reducing waiting times for surgery and preventing potential physical deterioration and psychological distress for this patient population, and eventually improving patient experience, safety and clinical effectiveness (an overview of anticipated impact is depicted in figure 1).

METHODS AND ANALYSIS

Study design

This mixed-methods cross-sectional study will analyse self-report ratings (eg, patient experience) and semi-structured responses (eg, patient preferences), collected from an online survey of individuals waiting for elective cardiac surgery. This study will be an explanatory concurrent design where qualitative data will be used to explain quantitative results.[24] Based on the commonly used Morse notation system for mixed-methods studies, this can be considered as a 'QUANT+qual' study with quantitative data as the core component (indicated by capital letters) and qualitative data as the supplemental component (the '+' indicates that the two components are collected simultaneously).[25] Specifically, quantitative and qualitative data will be collected at the same time (via the survey), and analysed separately on two different data sets. Integration of findings will be done with a side-by-side comparison in the discussion section wherein quantitative results will be confirmed/disconfirmed by the qualitative results. Any divergence between quantitative and qualitative findings will be followed by further exploration of data. If divergence persists this will be acknowledged as a limitation in the discussion.[26]

Sample

The sample will be selected using a convenience sampling method. Potential participants will be identified by screening waiting lists for elective cardiac surgery at each one of the four study sites. Based on the NHS guide for

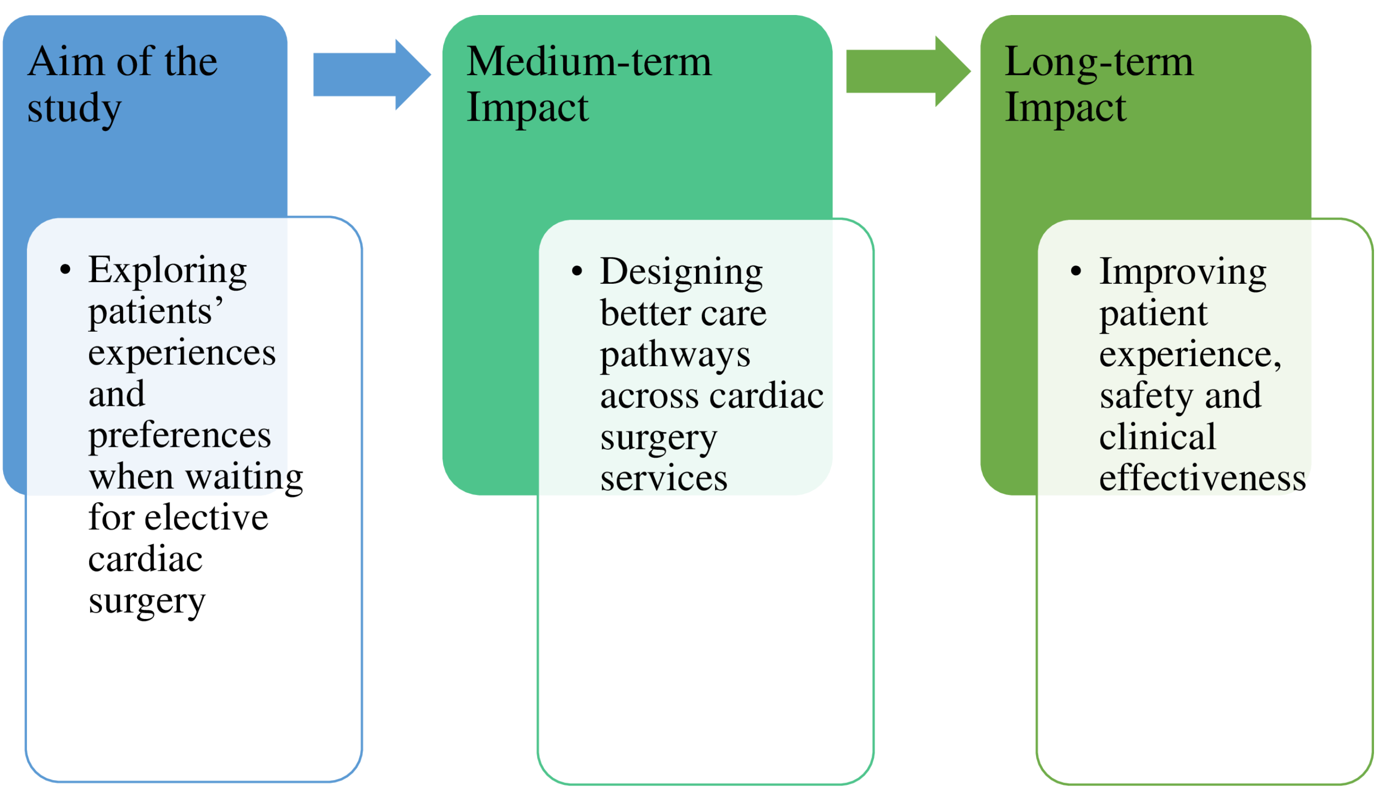

Figure 1 Overview of potential anticipated long-term study impacts. This figure describes the main aim of the study, and what the medium-term and long-term impact of the study will be.

surgical prioritisation,[27] waiting lists for elective surgery are grouped into four priority categories: P1 includes emergency patients (requiring surgery within 24 hours, therefore they are not on elective cardiac surgery waiting lists); P2 includes patients who require surgery within 4 weeks; P3 and P4 are patients who require surgery within 3 months, and between 3 and 6 months, respectively. Approximately 500 patients at any given time are on the waiting lists for elective cardiac surgery across the four participating hospitals in this study: Royal Brompton Hospital (RBH), Harefield Hospital (HH), St Thomas' Hospital (STH) and King's College Hospital (KCH). Although each site operates as a separate service, they are part of the King's Health Partner (KHP) Academic Health Science Centre, with collaboration and knowledge sharing across the sites on clinical matters including cardiac surgery. The waiting lists include patients in the P2, P3 and P4 categories whose most common diagnoses are coronary artery disease, aortic valve stenosis, mitral valve regurgitation and aortic aneurysm. Accordingly, the most commonly performed surgeries within the above categories are CABGs, aortic valve replacement, mitral valve repair or replacement and aortic aneurysm repair.

Screening and identification of and contact with potential eligible participants will be carried out by members of the healthcare team (eg, service/waiting list managers) at each participating site. Onboarding of the healthcare team was discussed and collaborations with researchers established during the preparation phase of the study.

Patients will be invited to participate in the study if they are: (1) on a waiting list for elective cardiac surgery across any of the four sites: RBH, HH, STH and KCH; (2) under surgical prioritisation categories P2, P3 or P4; and (3) at least 18 years old.

Patients will not be invited to participate if they are: (1) on a waiting list for elective cardiac surgery at sites other than RBH, HH, STH and KCH; and/or (2) do not have any recorded contact details in the form of mobile phone or home address.

Measures

The measure adopted in this study is an online survey that thanks to its anonymous format would encourage honest responses, and will ease practicalities around data collection with a speedy responses' submission from participants.

The survey consists of four parts: background information (Part A); impact of cardiac condition on life (Part B); experiences of cardiac surgery services (Part C); and preferences for how waiting lists and services could be improved (Part D). Following pre-testing, the time for completion of the survey is expected to be approximately 20 min.

The survey was developed collaboratively by researchers from the King's Improvement Science (KIS; a health and social care improvement research programme funded by KHP), cardiac surgeons and both internal (ie, recruited as part of the research team) and external patient and public involvement (PPI) groups with lived-experience of elective cardiac surgery. The survey was informed by the study research questions, evidence from the literature and relevant clinical and lived experience; namely, PPI members were deeply involved in defining areas of interests (ie, life domains potentially impacted by the cardiac condition, experience of the cardiac services/waiting list and potential preferences about how waiting lists could be improved) and questions about communications with the clinical team and surgery specific preferences (eg, willingness to wait, logistic) were suggested by both surgeons and PPI members. In its final version, the survey consists of 44 questions using multiple-choice and open-ended options (please see online supplemental material). A 5-point Likert scale is used to rate the frequency and importance of some items/questions, and text boxes (without a word limit) are provided for the open-ended questions. Based on feedback from the study PPI group (see below), all questions are optional, so participants can choose 'prefer not to answer' if they wish.

Given the high incidence of anxiety and depressive symptoms in patients suffering from CVDs,[19 21 22] the Patient Health Questionnaire (PHQ)-4 was included in Part B of the survey.[28] The PHQ-4 is a brief, validated, self-report scale that consists of four questions about anxiety and depressive symptoms which are rated on a 4-point Likert scale.

In line with standard reporting for electronic surveys, further details are provided in table 1 which follows the Checklist for Reporting Results of Internet E-Surveys (CHERRIES[29]).

Patient and public involvement

A total of 12 PPI members have been involved in this study since its inception. Two individuals with lived experience of cardiac surgery were recruited (via an advert circulated to PPI networks) to be part of the core research team. The two PPI research team members have and continue to attend regular research meetings and have provided input/feedback on study design, procedures and materials (eg, content of the participants information sheet (PIS) and survey, survey duration, number of invitations and modality to approach eligible participants). Also, one of the two PPI members attended the Research Ethics Committee (REC) meeting along with the chief investigator. The PIS and survey were also presented and discussed during monthly internal KIS PPI meetings, and during an ad hoc workshop with a PPI group from one of the partner hospitals in this study. All feedback from PPI members, including suggestions on layout, wording and acceptability of questions, were considered, balanced when there were disagreements and incorporated as appropriate in the documents. The survey was then trialled (without data collection) by four external PPI representatives who were not involved in any aspect of the survey's development. The aims of this trial were to estimate the duration of completion and obtain any final suggestions on the clarity and acceptability of the

Table 1 Checklist for Reporting Results of Internet E-Surveys (CHERRIES)

Item category	Checklist
Design	Mixed-methods (explanatory concurrent design) cross-sectional e-survey.
Ethics approval and informed consent process	**Ethics approval**: Research Ethics Committee approval ref number 23/EE/0138. **Informed consent**: in the initial section of the survey, participants will select if they consent to take part in the study. **Data protection**: survey responses will be collected directly via Research Electronic Data Capture (REDCap) and each response assigned a random numerical ID. Data will be maintained within King's College London Information Technology managed devices and supported digital locations, and will not be transferred to external institutions. Only the research team will access data. To mitigate de-identification at the results reporting stage, pseudonymisation and other data masking techniques will be applied.
Development and pre-testing	The survey was developed in an iterative fashion and close collaboration with people with lived experience, and cardiac surgeons. More details about survey development is reported in the text. The final version of the survey was trialled on a small group of patient and public involvement representatives who were not involved at any stage of the survey's development.
Recruitment process and description of the sample having access to the questionnaire	**Open/closed survey**: this is a closed survey that is sent only to eligible participants. **Contact mode**: a text and/or letter is sent by the healthcare team to eligible participants. **Advertising the survey**: the survey is advertised with a poster that is put up on the cardiac surgery services' waiting rooms. The poster contains information and contact details but not the link/QR code for participation.
Survey administration	**Type of e-survey**: this is an electronic survey accessible with a weblink and/or QR code sent via text/letter. **Context**: the survey is hosted on REDCap, an online GDPR*-compliant platform to capture data. **Mandatory/voluntary**: responding to the survey is voluntary; for each question, the participant has a 'prefer not to answer' option. **Incentives**: at the time of responses' submission, each participant has the possibility to enter a prize draw for a £20 voucher. **Time/date**: the survey has an 8-week window. Data is collected starting from October to November 2023. **Randomisation of items or questionnaire**: not applicable. **Adaptive questioning**: some questions use branching logic so that further questions are conditionally displayed based on previous responses. **Number of items**: 44 **Number of screens**: 12 **Completeness check**: before moving to the next page, an alert window will notify any incomplete field. A 'prefer not to answer' option is available for each question. Review step: participants are able to review, change and/or withdraw their answers during the life of the survey (8 weeks).
Response rates	**Unique site visitor and view rate**: not applicable. **Participation/recruitment rate**: ratio of participants who agreed to participate/started the survey over those who were invited to participate. **Completion rate**: ratio of participants who completed over those who agreed to participate.
Preventing multiple entries from the same individual	**Cookies used**: not applicable. REDCap allows the responder to complete the survey over a 30-min period, so if they close the page and then return, they will not be asked for a log-in Return code. **IP check**: not applicable. **Log file analysis**: not applicable. **Registration**: this is a closed survey wherein each entry has an associated link and Return code that can be used by the participant to return/change to their responses during the life of the survey.
Analysis	**Handling of incomplete questionnaires**: all submitted surveys will be analysed. **Questionnaires submitted with an atypical timestamp**: no time frame will be used as a cut-off point (to include/exclude surveys). **Statistical correction**: no statistical correction will be applied.

GDPR, General Data Protection Regulation; ID, identification number.

questions. Their feedback, submitted via an anonymous form, suggested feasibility and acceptance of the survey so that version was considered as the final version for the study. Involvement of PPI in the next stages of the study (ie, results, interpretation and dissemination) is planned. Once the study is completed, the short form of the Guidance for Reporting Involvement of Patients and the Public (GRIPP2-SF[30]) will be used to report PPI involvement across all stages of the research study.

Procedures

The survey will be launched online and open to responses for 8 weeks. During this time, participants will be able to both submit and withdraw responses as desired (see more

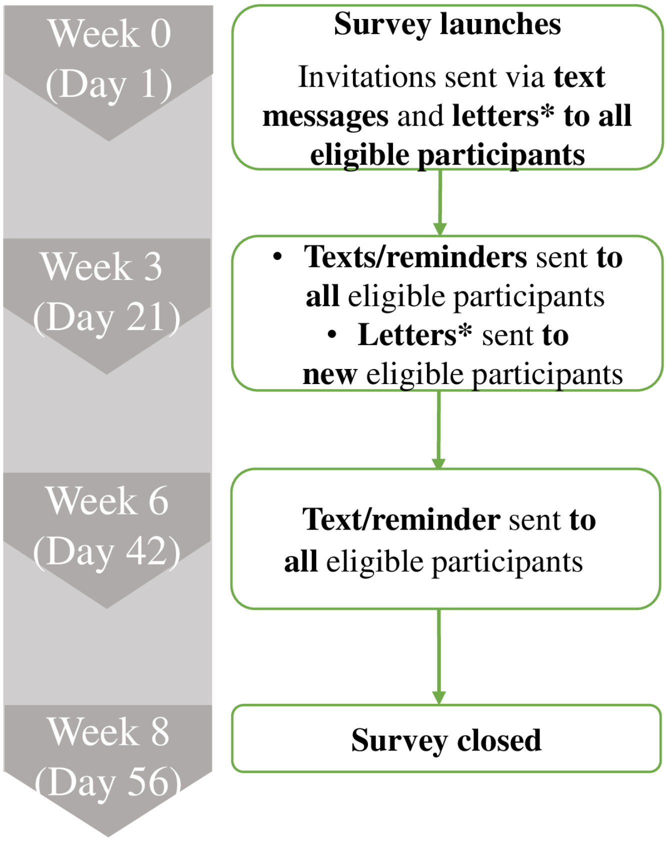

Figure 2 Survey delivery schedule. This figure shows the schedule of the survey, specifically the time points when invitations to eligible patients will be sent. *Only from those sites that will have capacity to send letters.

details about withdrawal procedures below). Each site will start sending invitations to patients on their waiting lists based on the healthcare team's availability. Data collection is planned to take place between late 2023 (October/November) and early 2024 (January/February).

Eligible participants will be invited to take part in the study by their cardiac service healthcare team. Invitations will be sent via text message and/or letter (depending on availability of contact information, and capacity of staff to send letters), which will contain a weblink (in the text messages) or web address and QR code (in the letters) to access the survey. Eligible participants will receive a maximum of four invitations to take part in the survey. However, we expect that only a proportion of eligible potential participants will receive all four invitations; this will depend on the contact details available for an eligible potential participant (eg, some may receive either the letter or the text message rather than both), the length of time they are on the waiting list during the 8-week window of the survey and the healthcare team's capacity to send letters.

Invitations will be sent according to the following schedule (figure 2): the first invitation will be sent via text message and/or letter (depending on what contact details are available) to all eligible potential participants when the survey launches online; after 3 weeks, a further round of invitations will be sent to all eligible potential participants: text messages will be sent to all individuals on the waiting lists, while invitation letters will only be sent to those who have been *newly added* to the cardiac surgery waiting lists since the launch of the survey; after three more weeks, a final text message invitation will be sent to all eligible patients, as a final reminder 2 weeks before the survey closes.

Study withdrawal will be allowed while the survey is live only (ie, during the 8-week window) due to the anonymous nature of the survey. If a participant decides to withdraw their responses while they are completing the survey, they will need to go back to one of the initial questions where they consented to take part in the study and select 'no'. By doing this, their data will be automatically deleted and not recorded as part of the study. Withdrawal will be possible after submitting the responses too (as far as this happens within the 8-week window). Each participant will be provided with a unique 'Return Code' when they take a break from the survey (ie, responses will be saved and participant can return to the survey at a later point) or when they submit the survey. This code will be needed to log back into the survey (via the weblink/QR code they were provided via text and/or letter) and select 'no' to the consent question as described above. The withdrawal procedure is detailed in the PIS.

Data management plan

Data collection

Participants' survey responses will be collected directly via Research Electronic Data Capture (REDCap), which is a browser-based data capturing software used to develop and host surveys.[31] Survey responses will be the only data source used in the study. Data confidentiality will be maintained by the automatic random assignment of an identification number (ID) to each survey response. Personal identifiers such as names, NHS number and post code will not be collected as part of the survey. After completing the survey, participants will be invited to follow a link to a separate REDCap form that will offer the opportunity to enter a thank you prize draw (for 1 of 50 × £20 e-vouchers). To enter the prize, participants will need to submit their email addresses. As these email addresses will be collected on a separate form, this data will not be linked to participants' survey responses. After the prize draw is completed, the list of email addresses will be destroyed on REDCap and on any King's College London (KCL) managed devices.

Data processing

Once data collection is completed, data will be exported from REDCap into Excel, and software packages for analysis: IBM SPSS Statistics (V.27) for quantitative analysis, and NVivo (V.12) for qualitative analysis. All collected data will be maintained within KCL, on KCL information technology (IT) managed devices and supported digital locations. No data will be transferred to external institutions.

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As data will include personal information, such as gender, ethnicity and sexual life, different techniques (eg, data masking, data aggregation), in line with the Information Commissioner's Office guidance, will be adopted to pseudonymise data. In the data set, data will be pseudonymised by the assignment of a progressive numerical ID (eg, 001, 002, 056, 345) which would not necessarily correspond to the one automatically assigned by REDCap at the time of data collection. When reporting results, participant quotes will be further pseudonymised with the attribution of a code to the responder (eg, R1, R2, R3…) and omission of any potentially identifiable information (eg, names, hospitals) provided by participants within free text responses to questions. When displaying numerical results in tables, any cell that might contain small values (eg, 1–5) will be suppressed or aggregated to avoid potential re-identification.

After study completion, in line with KCL policy, a data retention period of 5 years will be applied. The pseudonymised data set will be retained for 5 years on the King's Open Research Data System, a KCL data repository that provides long-term storage and access for data sets at the project-end and for supporting publications.

Data analysis plan

Sample characteristics (socio-demographic and clinical features collected in Part A of the survey) will be reported using descriptive statistics: frequencies and percentages for categorical variables, and means, medians, SD or IQRs for continuous variables, depending on their distribution. Results will be graphically reported and/or tabulated as appropriate.

To address aim 1 (ie, patients' experiences of how their cardiac condition impacts their life and how they experience being on a waiting list for cardiac surgery), data derived by Part B (ie, 'impact of cardiac condition on life') and Part C (ie, 'experience of cardiac surgery services') of the survey will be used. Descriptive statistics will be applied to numerical data to report participants' ratings on survey questions, including on the embedded PHQ-4 scale. Across all questions with Likert scales, a higher rating will correspond to a more negative experience.

For exploratory purposes, bivariable descriptive analyses will be carried out. Multivariable regression explorations with pairwise correlation will be conducted to assess the association between each of the items in Part B of the survey, and anxiety and depressive symptoms as measured by each of the four PHQ-4 items (ie, feeling nervous/anxious, worrying, feeling down, anhedonia/loss of pleasure). Specifically, items in Part B are ratings of the interference of cardiac condition on life domains and consist of the following ordinal variables: daily activities, social activities, studying or working, hobbies, sexual life and family/friend relationship. The strength of the correlation will be used as a measure of the magnitude of the association; values between 0.1 and 0.3 will be considered as weak correlation, between 0.4 and 0.6 as moderate and from 0.7 and above as strong/perfect.[28] Multivariable models will be adjusted for socio-demographic characteristics. Interpretation of these results will be complemented by results from qualitative analysis (see details in the Study design section above).

Textual data obtained via Parts B and C of the survey will be analysed using qualitative content analysis as described in detail below.

To address aim 2 (ie, patients' preferences in relation to the delivery of surgery and their views on improving the waiting process), data obtained from Part D (ie, preference on how the waiting list could be improved) of the survey will be used and quantitative (ie, descriptives statistics) and qualitative content analysis with an inductive approach will be applied.[32] The choice of the qualitative content analysis with an inductive approach is based on the lack of existing theories or frameworks to draw on in the field of cardiac patients' preferences.[32] Specifically, the content analysis would offer us the possibility to move flexibly, without a priori ideas, from specific units of analysis (eg, sentences) as they emerged from the text into higher order concept/themes.[33] The proposed qualitative analysis will be carried out separately for aims 1 and 2, as the data on which they will rely on are derived from different sections of the survey.

The inductive content analysis will follow different steps: (1) familiarisation and immersion in the data by repeatedly reading all data to achieve a sense of the whole; (2) two to three researchers will open code a portion (~10%) of the data (data units, basic unit of text representing codes, will be agreed based on impressions from coders); (3) based on familiarisation and initial independent open coding, a first draft of the coding framework will be created; (4) the first draft of the coding framework will then be applied to ~20% of the data by the researchers and inter-coder reliability (ICR) calculated with the Krippendorff alpha which allows for flexibility in number of coders and missing codes[34]; (5) based on the ICR (a value of at least 0.7 will be deemed satisfactory[34]), the coding framework may be revised or confirmed and then applied to the entire data set; (6) once the entire data has been coded, the codes can be sorted into higher order categories/subcategories (definitions for categories/subcategories are developed).[32 33 35]

Quantitative analysis, including the calculation of ICR values, will be done using SPSS. All qualitative analyses will be conducted using NVivo.

Patient and public members with relevant experience of CVD/waiting for elective cardiac surgery will be involved in data analysis, specifically in validating the coding framework to be applied in the qualitative analysis and in interpreting results obtained from both quantitative and qualitative analyses.

ETHICS AND DISSEMINATION

Ethical and safety considerations

This study was granted ethical approval from the East of England—Cambridge East REC (ref n. 23/EE/0138).

We acknowledge that completing the survey might be distressing for some participants, because it requires them to think about potentially unsettling experiences and emotions and to rate the presence of any anxiety and/or depressive symptoms. The inclusion of these questions is deemed necessary to address our research questions about the impact of patients' cardiac conditions on their lives and to explore any possible association between the level of clinical symptoms and any specific experiences and/or preferences for waiting list management. Survey questions were reviewed by PPI members with relevant lived experience to address any potential ethical issues. Important edits were made to some questions; for instance, it was thought to keep the focus on the understanding of the risk associated with the surgery rather than asking participants about what level of risk was specifically involved in their operation which could potentially introduce unnecessary anxiety. However, the anonymous nature of the survey will not allow us to follow-up with participants reporting any anxiety and/or depressive symptoms. To address this ethical issue, the PIS contains contact details of the chief investigator, and a list of relevant external organisations that can provide support to patients surrounding heart health and/or emotional distress and information about how to contact the Patient Advice and Liaison Service. To help cater for diverse needs, the list of external organisations includes six types of organisations, for example, the British Heart Foundation, Heart Valve Voice and The Samaritans, and highlights different avenues of support, for example, website information, online forums and telephone helplines. This signposting information has also been reviewed by patient and public members and is deemed to be acceptable. The signposting information will also be included at the end of the survey. Beyond ethical consideration about data safety and processing that are described in an earlier section above, there are no other major ethical, safety, legal or management issues expected.

Dissemination

Dissemination of the study results will be targeted to different types of audiences: academics/researchers, patients and carers, public members, clinicians, managers and other stakeholders.

It is intended that two main scientific publications to Open Access peer-reviewed journals will originate from this study: the study protocol (current work), and results from the survey. Study findings will also be submitted for presentation to relevant conferences and events.

Study findings will be presented at meetings which will include cross-disciplinary events organised in collaboration with clinical leads from the study. The format of the event(s) (online or in-person, participation in pre-existing events) will be decided based on requirements and resources. To provide a balanced communication between researchers on one side, and patients, carers and the general public members on the other, findings will be disseminated through a PPI commentary (eg, reflections on findings and/or experience about being part of the project) accompanying the manuscript about main study results, a blog on the KIS website, a PPI webinar and linking up with internal and external PPI groups, networks and third-sector organisations.

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REFERENCES

1 Bhatnagar P, Wickramasinghe K, Williams J, *et al*. The epidemiology of cardiovascular disease in the UK 2014. *Heart* 2015;101:1182–9.
2 Guidance health matters: preventing cardiovascular disease. 2019. Available: https://www.gov.uk/government/publications/

health-matters-preventing-cardiovascular-disease/health-matters-preventing-cardiovascular-disease
3 The untold heartbreak. 2021. Available: https://www.bhf.org.uk/what-we-do/news-from-the-bhf/news-archive/2021/august/pandemic-backlog-heart-care-five-years-recover [Accessed 9 Aug 2021].
4 NHS England. Consultant-led referral to treatment waiting times data 2022-23. 2023. Available: https://www.england.nhs.uk/statistics/statistical-work-areas/rtt-waiting-times/rtt-data-2022-23/
5 Yates MT, Balmforth D, Kirmani BH, *et al*. A multi-centre prospective cohort study of patients on the elective waiting list for cardiac surgery during the COVID-19 pandemic. *J R Soc Med* 2022;115:348–53.
6 Head SJ, da Costa BR, Beumer B, *et al*. Adverse events while awaiting myocardial revascularization: a systematic review and meta-analysis. *Eur J Cardiothorac Surg* 2017;52:206–17.
7 Arthur HM, Daniels C, McKelvie R, *et al*. Effect of a preoperative intervention on preoperative and postoperative outcomes in low-risk patients awaiting elective coronary artery bypass graft surgery. A randomized, controlled trial. *Ann Intern Med* 2000;133:253–62.
8 McCormick KM, Naimark BJ, Tate RB. Uncertainty, symptom distress, anxiety, and functional status in patients awaiting coronary artery bypass surgery. *Heart Lung* 2006;35:34–45.
9 Abate SM, Chekol YA, Basu B. Global prevalence and determinants of preoperative anxiety among surgical patients: a systematic review and meta-analysis. *Int J Surg Open* 2020;25:6–16.
10 Navarro-García MA, Marín-Fernández B, de Carlos-Alegre V, *et al*. Preoperative mood disorders in patients undergoing cardiac surgery: risk factors and postoperative morbidity in the intensive care unit. *Rev Esp Cardiol* 2011;64:1005–10.
11 Hernández-Palazón J, Fuentes-García D, Falcón-Araña L, *et al*. Assessment of preoperative anxiety in cardiac surgery patients lacking a history of anxiety: contributing factors and postoperative morbidity. *J Cardiothorac Vasc Anesth* 2018;32:236–44.
12 Gallagher R, McKinley S. Stressors and anxiety in patients undergoing coronary artery bypass surgery. *Am J Crit Care* 2007;16:248–57.
13 Koivula M, Paunonen-Ilmonen M, Tarkka MT, *et al*. Fear and anxiety in patients awaiting coronary artery bypass grafting. *Heart Lung* 2001;30:302–11.
14 Gagliardi AR, Yip CYY, Irish J, *et al*. The psychological burden of waiting for procedures and patient-centred strategies that could support the mental health of wait-listed patients and caregivers during the COVID-19 pandemic: a scoping review. *Health Expect* 2021;24:978–90.
15 Carr T, Teucher U, Casson AG. Waiting for scheduled surgery: a complex patient experience. *J Health Psychol* 2017;22:290–301.
16 Carr T, Teucher UC, Casson AG. Time while waiting: patients' experiences of scheduled surgery. *Qual Health Res* 2014;24:1673–85.
17 Cesena FHY, Favarato D, César LAM, *et al*. Cardiac complications during waiting for elective coronary artery bypass graft surgery: incidence, temporal distribution and predictive factors. *Eur J Cardiothorac Surg* 2004;25:196–202.
18 Ivarsson B, Larsson S, Sjöberg T. Patients' experiences of support while waiting for cardiac surgery. A critical incident technique analysis. *Eur J Cardiovasc Nurs* 2004;3:183–91.
19 Shibeshi WA, Young-Xu Y, Blatt CM. Anxiety worsens prognosis in patients with coronary artery disease. *J Am Coll Cardiol* 2007;49:2021–7.
20 Lett HS, Blumenthal JA, Babyak MA, *et al*. Depression as a risk factor for coronary artery disease: evidence, mechanisms, and treatment. *Psychosom Med* 2004;66:305–15.
21 Rutledge T, Linke SE, Krantz DS, *et al*. Comorbid depression and anxiety symptoms as predictors of cardiovascular events: results from the NHLBI-sponsored women's ischemia syndrome evaluation (WISE) study. *Psychosom Med* 2009;71:958–64.
22 Goodwin RD, Davidson KW, Keyes K. Mental disorders and cardiovascular disease among adults in the United States. *J Psychiatr Res* 2009;43:239–46.
23 Kreindler SA. Policy strategies to reduce waits for elective care: a synthesis of international evidence. *Br Med Bull* 2010;95:7–32.
24 Schoonenboom J, Johnson RB. How to construct a mixed methods research design. *Kolner Z Soz Sozpsychol* 2017;69(Suppl 2):107–31.
25 Morse JM, Niehaus L. *Mixed Method Design: Principles and Procedures*. Walnut Creek: Left Coast Press, 2009.
26 Creswell JaPC VL. *Designing and Conducting Mixed Methods Research: 2nd ed*. Los Angeles: Sage, 2011.
27 NHS England. Clinical validation of surgical waiting lists: framework and support tools. 2021. Available: https://www.england.nhs.uk/coronavirus/documents/clinical-validation-of-surgical-waiting-lists-framework-and-support-tools/#appendix-d
28 Kroenke K, Spitzer RL, Williams JBW, *et al*. An ultra-brief screening scale for anxiety and depression: the PHQ-4. *Psychosomatics* 2009;50:613–21.
29 Eysenbach G. Improving the quality of web surveys: the checklist for reporting results of internet E-surveys (CHERRIES). *J Med Internet Res* 2004;6:e34.
30 Staniszewska S, Brett J, Simera I, *et al*. Gripp2 reporting checklists: tools to improve reporting of patient and public involvement in research. *BMJ* 2017;358:j3453.
31 Harris PA, Taylor R, Minor BL, *et al*. The Redcap consortium: building an international community of software platform partners. *J Biomed Inform* 2019;95:103208.
32 Hsieh HF, Shannon SE. Three approaches to qualitative content analysis. *Qual Health Res* 2005;15:1277–88.
33 Elo S, Kyngäs H. The qualitative content analysis process. *J Adv Nurs* 2008;62:107–15.
34 Nili A, Tate M, Barros A, *et al*. An approach for selecting and using a method of inter-coder reliability in information management research. *Int J Inf Manag* 2020;54:102154.
35 O'Connor C, Joffe H. Intercoder reliability in qualitative research: debates and practical guidelines. Int J Qual Methods 2020;19:160940691989922.